Problem 5 | Problems T3_Diffusion.pdf #5 | Fick's 2nd Law + Arrhenius Coupled

Cementation Temperature for Steel 1010 in 8 Hours

Original statement · Problems T3_Diffusion.pdf

5.- A 1010 steel, that contains $0.10\ \text{wt\%}\ \text{C}$ *is subjected to cementation process, in order to obtain effective* C *composition of* $0.30\ \text{wt\%}$ *at a depth of* $1.16\ \text{mm}$ *below the surface. Assume that the maximum* C *solubility in the steel surface is reached at moment that the carburizing process is initiated, generating a* C *concentration at the surface of* $1.31\ \text{wt\%}$*. Determine the required cementation temperature for the process to be successfully carried out within an 8 hour time span. Data:* $D_0(\text{C in Fe}) = 16.2\ \text{mm}^2/\text{s}$*;* $E_D = 137\,800\ \text{J/mol}$*;* $R = 8.314\ \text{J/mol}\cdot\text{K}$*.*

Official Solution: T = 1175 K = 902 °C

PHASE 1 · HYPOTHESES & PARAMETERS

- **Coupled Transient Diffusion:** Combines semi-infinite error function diffusion with Arrhenius thermal kinetics [Session 5 Slides 20, 26].
- **Parameters:** $C_0 = 0.10\ \text{wt\%}$, $C_s = 1.31\ \text{wt\%}$, $C_x = 0.30\ \text{wt\%}$ $x = 1.16\ \text{mm} = 1.16 \times 10^{-3}\ \text{m}$ $t = 8\ \text{h} = 8 \times 3600\ \text{s} = 28\,800\ \text{s}$ $D_0 = 16.2\ \text{mm}^2/\text{s} = 16.2 \times 10^{-6}\ \text{m}^2/\text{s}$ $E_D = 137\,800\ \text{J/mol}$, $R = 8.314\ \text{J/mol}\cdot\text{K}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

1. Evaluate the concentration ratio $\text{erf}(z) = \frac{C_s - C_x}{C_s - C_0}$.
2. Determine z via linear interpolation in the official table [Slide 21].
3. Calculate the required diffusivity: $D = \frac{x^2}{4 \cdot z^2 \cdot t}$.
4. Invert the Arrhenius equation to calculate the processing temperature T:

$$T = \frac{-E_D}{R \cdot \ln(D/D_0)}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Error Function Value:

$$\operatorname{erf}(z) = \frac{1.31 - 0.30}{1.31 - 0.10} = \frac{1.01}{1.21} \approx 0.83471074$$

2. Linear Interpolation in Error Function Table [Slide 21]:

- For $z_1 = 0.95 \implies \operatorname{erf}(z_1) = 0.8209$
- For $z_2 = 1.00 \implies \operatorname{erf}(z_2) = 0.8427$

$$\frac{z - 0.95}{1.00 - 0.95} = \frac{0.83471074 - 0.8209}{0.8427 - 0.8209} \implies \frac{z - 0.95}{0.05} = \frac{0.01381074}{0.0218} \approx 0.633520$$

$$z = 0.95 + 0.05 \times 0.633520 = 0.981676 \implies z^2 = (0.981676)^2 \approx 0.963688$$

3. Diffusivity Calculation:

$$x^2 = (1.16 \times 10^{-3}\ \mathrm{m})^2 = 1.3456 \times 10^{-6}\ \mathrm{m}^2$$

$$4 \cdot z^2 \cdot t = 4 \times 0.963688 \times 28\,800\ \mathrm{s} = 111\,016.86\ \mathrm{s}$$

$$D = \frac{1.3456 \times 10^{-6}\ \mathrm{m}^2}{111\,016.86\ \mathrm{s}} \approx \mathbf{1.21207 \times 10^{-11}}\ \mathrm{m^2/s}$$

4. Inverting Arrhenius for Temperature T:

$$\frac{D}{D_0} = \frac{1.21207 \times 10^{-11}\ \mathrm{m^2/s}}{16.2 \times 10^{-6}\ \mathrm{m^2/s}} = 7.48191 \times 10^{-7} \implies \ln\left(\frac{D}{D_0}\right) = \ln(7.48191 \times 10^{-7}) = 2.01249 - 16.11810 = -14.10561$$

$$T = \frac{-137\,800\ \mathrm{J/mol}}{(8.314\ \mathrm{J/mol \cdot K}) \times (-14.10561)} = \frac{137\,800}{(8.314)(14.10561)} = \frac{137\,800}{117.275} \approx \mathbf{1175.02}\ \mathrm{K} \approx \mathbf{1175}\ \mathrm{K}$$

$$T = 1175.02 - 273.15 = \mathbf{901.87}^{\circ}\mathrm{C} \approx \mathbf{902}^{\circ}\mathrm{C}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Sensitivity to Temperature: To reach $0.30\ \mathrm{wt}\%$ at $1.16\ \mathrm{mm}$ in $8\ \mathrm{h}$ the steel must be carburized at $902^{\circ}\mathrm{C}$. Because D grows exponentially with T, a lower temperature would need a much longer time and a higher one a shorter time [Slide 26].